POLITECNICO
MILANO 1863

BASI DI DATI 1
Anno Accademico 2023-2024
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Appello 19/01/2024 **DURATA: 2 ore**

Part 1 – QUERY LANGUAGES

WEBSITE(SiteCode, Name, Admin, URL)
PAGE(PageCode, SiteCode, Title, Content, Author, PubDate, Category*)
COMMENT(PageCode, SiteCode, ID, Date, User, Text)
USER(Username, Email, Country)

In the above schema we want to store some useful information about a network of Web sites. Specifically, for the **pages** of the sites we want to store: a 5-character alphanumeric PageCode, the integer SiteCode, the title, the content (which is stored as text of arbitrary length), the user who created the page, the date of publication, and the category (which, if known, can store the values "Technology," "Art," or "Cooking"). All users who register on the network (including site administrators and page creators) can post comments on the pages of all sites.

Exercise 1 – SQL (13pt)

A. Define the PAGE schema, including referential integrity constraints. (2pts)
B. Find for each site the number of pages without any comments written by Italian users. (4pts)
C. Considering only comments on uncategorized pages created in 2023, find the country whose users produced more comments than the average number of comments for those pages. (4pts)
D. Express the constraint that requires that the comments have a date after the creation of the page to which they are associated. An exception must be made by the administrators of the site of which the page is part, who can comment on a page even before its publication. (3pts)

Exercise 2 – Formal languages (6pt)

Express in optimized relational algebra and Datalog the following query:

Find the countries of users who have commented on more than one page.

PART 2 – DATABASE DESIGN

Exercise 3 – Theory (1pt)

Define referential integrity management policies.

Exercise 4 - Conceptual and Logical Design (13pt)

- *The design is evaluated for completeness, correctness, readability, minimality, and self-explanation.*
- *Remember to specify the identifiers of all entities and the cardinality of all associations.*

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